

MCS-014 - System Analysis and Design

Most Repeated Questions (2010-2023)

IGNOU MCA/BCA

1. Data Flow Diagrams (DFDs)

1. Draw DFDs up to Level-2 for a Library Management System.
2. Draw DFDs up to Level-2 for a Hospital Information System.
3. Draw DFDs up to Level-2 for an Online Examination System.
4. Draw DFDs up to Level-2 for a Railway Reservation System.
5. Draw DFDs up to Level-2 for a Hotel Management System.
6. Draw DFDs up to Level-2 for a Student Information System.
7. Draw DFDs up to Level-2 for a Grade Card Generation System.

2. Entity-Relationship Diagrams (ERDs)

8. Draw an ERD for a Library Management System.
9. Draw an ERD for a Hospital Information System.
10. Draw an ERD for a Railway Reservation System.
11. Draw an ERD for a Hotel Management System.
12. Draw an ERD for a Visitor Management System.
13. Draw an ERD for an Online Shopping System.

3. SDLC Models & Phases

14. Explain the Waterfall Model with a diagram.
15. Explain the Spiral Model with a diagram.
16. Explain the Prototyping Model with a diagram.
17. Compare Waterfall Model and Spiral Model.

18. How can Waterfall and Prototyping be integrated into the Spiral Model?

19. Describe the phases of SDLC in detail.

20. What is the importance of SDLC in system development?

4. Software Requirements Specification (SRS)

21. Prepare an SRS for a Library Management System.

22. Prepare an SRS for an Online Examination System.

23. Prepare an SRS for a Hospital Information System.

24. Prepare an SRS for a Hotel Management System.

25. Prepare an SRS for an Online Shopping System.

5. Systems Analyst

26. Describe the role of a Systems Analyst in system development.

27. What are the skills and attributes of a good Systems Analyst?

6. Management Information Systems (MIS)

28. What is MIS? Explain its functions and characteristics.

29. Discuss how MIS helps in management control and strategic planning.

30. Differentiate between MIS and DSS.

7. Decision Support Systems (DSS)

31. What is a Decision Support System?

32. Describe the components of DSS with a diagram.

33. What are the characteristics of DSS?

8. CASE Tools

34. What are CASE tools?

- 35. Explain the components of CASE tools with a diagram.
- 36. What are the advantages and disadvantages of CASE tools?
- 37. Explain the role of CASE tools in technical and managerial aspects of software development.

9. Fact-Finding Techniques

- 38. Describe fact-finding techniques such as interviews, questionnaires, and observations.
- 39. What are the advantages and disadvantages of different fact-finding methods?
- 40. What is Joint Application Development (JAD)? Explain its process and participants.

10. Coupling and Cohesion

- 41. What is coupling? Explain its types.
- 42. What is cohesion? Explain its types.
- 43. Differentiate between coupling and cohesion.
- 44. Discuss the problems of high coupling between modules.

11. Software Maintenance

- 45. What is software maintenance?
- 46. Explain the types of maintenance: corrective, adaptive, perfective, preventive.
- 47. Why is software maintenance important? What issues are involved?

12. Cost-Benefit Analysis

- 48. What is cost-benefit analysis?
- 49. Explain any method of performing cost-benefit analysis with an example.

13. Database Design

- 50. Explain database models: relational, hierarchical, network.
- 51. What are the guidelines for designing databases?

52. Explain normalization with an example.

53. What is denormalization? Explain with example.

14. Forms and Reports

54. What are forms and reports?

55. Explain the criteria for designing forms and reports.

56. What are the steps in form design?

15. Expert Systems

57. What is an Expert System?

58. Describe the components of an Expert System.

59. What are the advantages and disadvantages of Expert Systems?

16. Other Important Areas

60. What is a Data Dictionary? What are its uses?

61. Explain Risk Analysis and Management.

62. Differentiate between Function-Oriented and Object-Oriented Design.

63. What is Information Security Architecture?

64. Explain User Interface Design guidelines.

65. What are Real-Time Systems and Distributed Systems?

66. What is Software Re-engineering and Reverse Engineering?

67. Explain Audit of Transactions.

68. What are the types of system testing? (Black Box, White Box, etc.)

Tip: Focus first on DFDs, ERDs, SRS, SDLC Models, MIS, DSS, CASE Tools, and Systems Analyst role.

These are most frequent.